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Research Article

A feast of aphids: assessing the detectability of prey aphid DNA inside ladybeetle predators using taxon-specific metabarcoding

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Abstract

Understanding insect predator–prey interactions is critical for crop protection, particularly within biological control and integrated pest management frameworks. Ladybeetles (Coleoptera, Coccinellidae) are widely used biocontrol predators of aphids (Hemiptera, Aphididae), which are among the most economically damaging pests of grains worldwide. However, post-release dietary assessment of ladybeetles remains challenging because of their polyphagous feeding behaviour. DNA metabarcoding provides a powerful molecular approach for resolving arthropod community composition and trophic interactions through simultaneous amplification and identification of multiple DNA fragments from mixed samples. Here, we applied a widely used insect surveillance metabarcoding workflow, modified through development and implementation of aphid-specific primers, to enhance targeted detection of aphid prey consumed by ladybeetles. We evaluated performance of this prey-targeted metabarcoding approach for dietary analysis and quantified aphid DNA detectability under controlled feeding and starvation regimes. We hypothesised that aphid DNA could be reliably recovered from whole ladybeetle samples using standard surveillance metabarcoding protocols, enabling detection of prey taxa present exclusively as gut contents, which may impact surveillance and diagnostics programs. Aphid DNA from multiple species was detected up to seventy-two hours post-ingestion and up to one hundred hours following cessation of *ad libitum* feeding, although detectability varied among aphid species and pre-feeding starvation treatments. These results demonstrate that prey-targeted metabarcoding increases sensitivity for dietary inference in predatory insects and provides a robust framework for detecting cryptic trophic interactions. Our findings further indicate that unintended prey detection may be common in arthropod metabarcoding datasets, with important implications for interpretation in biosurveillance and biosecurity monitoring.

Key words: Biological control, diagnostics, integrated pest management surveillance

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Introduction

Animal feeding behaviours shape food-web ecology and the interactions between predatory insects and their food (Gagnon et al. 2011; Casey et al. 2019; Masonick et al. 2019; Wilson Rankin et al. 2023). Understanding the interactions between predator and prey, as well as between herbivorous insects and their host plants, is critical for agriculture, pest management, and plant protection (Moorhouse-Gann et al. 2018; Zhu et al. 2019; Avanesyan and Lamp 2020; Cao and Dietrich 2021; Saqib et al. 2022; Berteloot et al. 2024). As a result, effective biocontrol of pests requires accurate and efficient assessments of predator-prey interactions, including both detecting pests when they are present in an area, and identifying if multiple predators are consuming those pests (Rosenheim and Harmon 2006; Rosenheim et al. 1993, 1999). Classical surveillance methods involving observation in the field and/or the collection and identification of pests are often laborious and challenging, and they are also less effective for visually cryptic or rare species (Heimpel et al. 1997; Rosenheim et al. 1999; Martoni et al. 2024). Such methods are also ineffective for monitoring efficacy of biocontrol agents (Greenstone et al. 2014; Hodgetts et al. 2016; Zhu et al. 2019).

High-throughput sequencing (HTS) metabarcoding provides a rapid approach to surveillance and biosecurity (Piper et al. 2019). This method can use general "universal" PCR primers to identify multiple arthropods from a bulk sample, and it has been widely used in Australia to identify pest and beneficial insects across different industries (Piper et al. 2019; Batovska et al. 2021; Martoni et al. 2023a, 2023b). Alternatively, by using target-specific primers, this method can explore the gut content or frass of a variety of organisms to assess their diet (Casey et al. 2019; Snider et al. 2022; Bucklin et al. 2024) or confirm that biocontrol species are, in fact, consuming their target pests (Jachowicz and Kramer 2025), in a similar way to a targeted quantitative PCR approach (Bucklin et al. 2024).

Metabarcoding of gut contents has been widely used to study the diets of insect predators, including ladybirds (Rondoni et al. 2018). However, the power of this technique may be limited by DNA detectability periods, as prey DNA within the predator tends to break down as it is digested (Weber and Lundgren 2009; Greenstone et al. 2014). More importantly, detectability periods differ depending on both the predator and prey species tested (Greenstone et al. 2014). Additionally, relative prey abundances may impact detectability: if there are few prey, there will be less DNA to detect, and if one prey species is highly abundant and hence more detectable, rarer species may be even harder to detect. Moreover, absolute and relative prey abundances may vary within or between cropping seasons, and may even be interlinked with predator feeding rates, starvation periods, and metabolic rates (Greenstone et al. 2014; Martoni et al. 2023a). Hence, these factors need to be considered when management requires a clear understanding of total prey presence and the timing of deployment of biological control agents.

The global grains industry is threatened by a range of non-native aphid species (Dedryver et al. 2010). In Australia, this includes the Russian wheat aphid *Diuraphis noxia* (Mordvilko in Kurdjumov, 1913) (Yazdani et al. 2018) and the corn leaf aphid *Rhopalosiphum maidis* (Fitch, 1856) (Da Barro 1991). Efforts to control populations of these two pest species through integrated pest

management requires surveillance to detect both their presence in agricultural sites to know where to deploy biocontrol agents (Martoni et al. 2023), as well as confirm their consumption by biocontrol agents, such as ladybeetles. Insect metabarcoding has been broadly adopted in agricultural surveillance and diagnostic projects (Hawthorne et al. 2025) to taxonomically identify members of the arthropod communities collected by active and passive trapping (Batovska et al. 2021; Martoni et al. 2023a; Martoni et al. 2024), including aphids in the grains industry (Martoni et al. 2023a). However, when comparing the results of metabarcoding assessments with visual examination of the arthropod sample, some studies have shown incongruencies where DNA was identified as belonging to species, but no physical insect specimen was present in the sample (Martoni et al. 2023a). Whether these incongruencies are due to ingested prey DNA, environmental DNA contamination, or both, remains unclear.

In this study, we assessed the utility of a broadly adopted insect metabarcoding approach (Martoni et al. 2023a, 2023b), slightly modified to use novel aphid-specific primers, to detect predation of different aphid species by ladybeetles, using DNA extracts of both whole predator bodies and frass. We hypothesized that aphid DNA could be detected from ladybeetle samples, leading to the detection of prey species that are no longer physically present in the bulk sample as distinct specimens. We also hypothesized that detectability of aphid DNA would reduce with increased time since feeding, and that detectability periods would be impacted by longer starvation periods before aphid consumption.

Methods

Insect collections and housing

Experimental aphid species were either sourced from the wild (*Uroleucon* spp. and cabbage aphids, *Brevicoryne brassicae*; Suppl. material 3) or from laboratory-reared colonies (corn leaf aphids, *Rhopalosiphum maidis*; Russian wheat aphids, *Diuraphis noxia*; and bird cherry-oat aphids, *Rhopalosiphum padi*; Suppl. material 3). Colonies of *R. maidis* and *D. noxia* were maintained on caged barley (*Hordeum vulgare*) plants grown from seeds in the laboratory at 28 °C under a 12:12 light:dark cycle for four weeks prior to feeding experiments. *Uroleucon* spp. and *B. brassicae* were collected from sow thistle (*Sonchus* sp.) and field mustard (*Brassica rapa*) plants, respectively, on the La Trobe University Bundoora campus, Victoria, Australia, an hour prior to feeding experiments. As positive controls for the metabarcoding experiments, we used a subset of aphids from each of the sources listed above, as well as California maple aphids (*Periphyllus californiensis*) collected from a Japanese maple tree (*Acer palmatum*) in the Horsham Botanic Gardens, Victoria, Australia. We also collected wild psyllids from an *Acacia* tree on the same campus to be used as non-target 'feeder' insects. For feeding experiments, we used spotted amber ladybeetles (*Hippodamia variegata*), a known predator in Australian grain ecosystems (Martoni et al. 2023a), that can be commercially sourced as eggs and adults (Biological Services, Loxton, Australia). We also collected large spotted ladybeetle (*Harmonia conformis*) larvae and adults from garden plants on the campus of La Trobe University, Bundoora, Victoria, Australia (37°43'18.8"S, 145°03'03.3"E) for additional comparisons.

Short starvation experiment

The first feeding experiment simulated ladybeetles gorging during a period of high pest abundance, followed by a short starvation period (17 hours) and then exposure to low abundances of additional prey (Fig. 1A). Here, we used ladybeetles (*H. variegata*) that were housed in mesh cages containing barley plants and were provided with *ad libitum R. maidis* (chosen because it was our most abundant aphid colony) as prey for nine days, and then starved for 17 hours prior to the experiment. Ladybeetles were stored individually in sterile petri dishes, and five aphids (either all *Uroleucon* spp. or *D. noxia* – we used different species to allow measurements of detectability after a specific feeding event, as *R. maidis* was likely still present in the ladybeetles guts) were introduced to each dish. Ladybeetles were given 1.5 hours to feed, and we recorded the number of aphids consumed, after which the remaining aphids were removed from the petri dish. We then collected ladybeetles into sterile tubes containing 100% ethanol at the following time points after feeding: 0 hr (i.e. immediately after the feeding period concluded), 1 hr, 2 hr, 4 hr, and 72 hr. We were unable to include timepoints between 4 and 72 hours due to logistical constraints. To provide controls for sequencing, a subset of ladybeetles (which had fed on *R. maidis* in the mesh cage, but were not fed on *Uroleucon* spp. or *D. noxia*) and samples from each aphid species (not exposed to ladybeetles) were selected at random, at the same time that feeding trials began, into 100% ethanol (in separate sterile vials).

Long starvation experiment

The second feeding experiment simulated the start or end of the season, with pest prey present in low abundance, to test the effects of a longer starvation period (96 hours) and to reduce the effect of dominant prey on DNA detectability (Fig. 1B). Here, we transferred all ~70 adult ladybeetles (*H. variegata*) from our colony evenly across two aphid-free barley cages (30 cm × 30 cm × 70 cm), where they were left for 96 hours without food prior to the feeding experiment. As in the Short starvation experiment, each ladybeetle was transferred to sterile petri dishes and provided with five aphids (either *R. maidis* or *B. brassicae* – we used these species as they were the most readily available at the time of the experiment). The ladybeetles were left for 1.5 hours to feed. After this feeding period, we recorded the number of aphids consumed, removed remaining aphids, and then transferred ladybeetles to 100% ethanol (in separate sterile vials) at one the following time points: 0 hr, 1 hr, 2 hr, 4 hr, 24 hr, 48 hr, and 96 hr. Different time points were used compared with the Short starvation experiment again due to logistical constraints. To maintain biologically realistic metabolic rates, ladybeetles in the 48 hr and 96 hr collection groups were fed two psyllids after 24 hr, and, for the 96 hr treatment, an additional two psyllids were provided after 72 hr (Weber and Lundgren 2009). As above, a subset of ladybeetles, aphids, and psyllids were transferred, from colonies or wild collections as applicable, into 100% ethanol (in separate sterile vials) at the beginning of the feeding period as controls for sequencing.

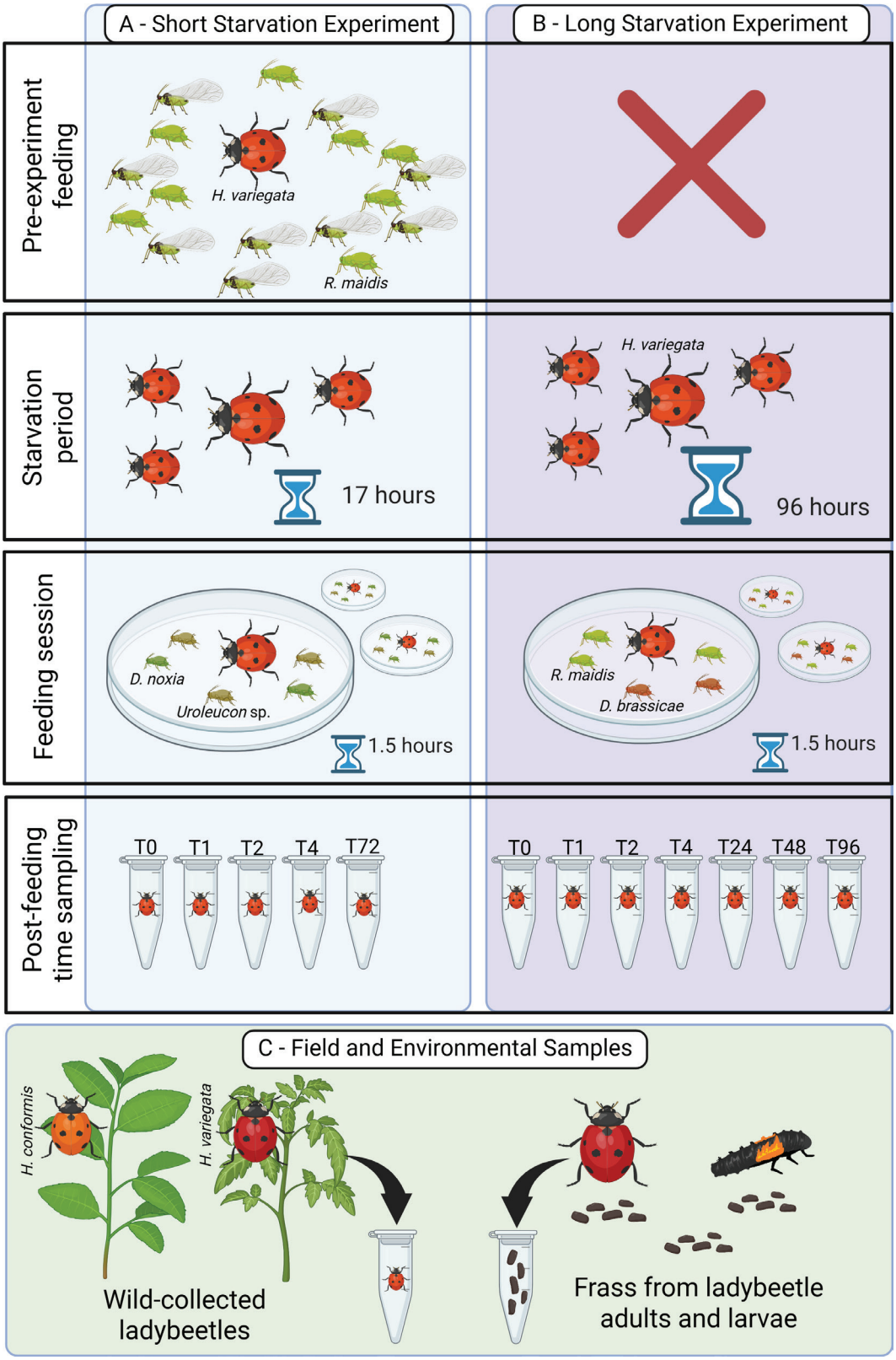


Figure 1. Design of the two experiments conducted in this study. The Short starvation experiment (**A**) and the Long starvation experiment (**B**) involved: adult *H. variegata* as the predator; a period of pre-feeding on *R. maidis* (**A**), or its absence (**B**); a starvation period of 17 or 96 hours; individual beetles exposed for 1.5 hours to 5 individuals of *D. noxia* or *Uroleucon* sp. (**A**), or *R. maidis* or *B. brassicae* (**B**), for feeding; then post-feeding sampling for metabarcoding at 0, 1, 2, 4 and 72 hours (**A**), or 0, 1, 2, 4, 24, 48 and 96 hours (**B**). Field-collected *H. variegata* and *H. conformis*, as well as frass from adult and larval *H. variegata*, were also sampled for metabarcoding (**C**). Created in BioRender. Martoni, F. (2026) https://BioRender.com/a8ek20r.

Field and environmental samples

To complement our experimental feeding trials, we sampled two ladybeetle species (*H. conformis* and *H. variegata*) collected directly from the field, and we analysed frass collected from ladybeetles during the Short starvation experiment (Fig. 1C). As a pilot trial, we collected frass (transferred into ATL buffer; see below) from three *H. variegata* larvae (from the barley cage colony) after feeding, and from two *H. conformis* (field-collected) and one *H. variegata* (lab colony) adults before and after feeding them field-caught aphids overnight. To minimise the amount of frass derived from food consumed in the pre-feeding period (Davidson and Evans 2011), larvae were moved to a new petri dish after feeding, where they remained for 24 hours, at which point the larvae and any frass were transferred to 100% ethanol in separate vials.

DNA extractions and library preparation

We performed destructive DNA extractions on most specimens, and non-destructive DNA extractions on some specimens, using the DNeasy Blood and Tissue kit (Qiagen, Germany) with an overnight incubation period (~17 hours) at 56 °C as previously described (Martoni et al. 2021). Initial pilots of non-destructive extractions from beetles produced inconsistent PCR bands, so destructive extractions were primarily used for our experiments. For destructive extractions, we dissected the ladybeetle into three parts (head and thorax together, and abdomen in two halves) and immediately transferred all three together into ATL buffer. These dissections took place in a biosafety cabinet with a new scalpel blade and new pair of gloves for each dissection and also included DNA extraction blank controls for each batch of samples processed. For comparison, we performed additional non-destructive extractions, without the dissection step (i.e. extracted from whole specimens), for some ladybeetles, as well as all frass, aphid, and psyllid specimens. For both extraction methods, we submerged specimens in 180 µL of ATL buffer and 20 µL of proteinase K at 56 °C overnight.

We purified the 200 µL lysate per sample on DNeasy spin columns, following the manufacturer's instructions. After DNA extraction, specimens processed non-destructively were preserved in high grade ethanol (100%) for further morphological examination, if required. We used a two-step PCR protocol, based on previous work (Martoni et al. 2023a, 2023b), to amplify target DNA and index sequences for Illumina sequencing. First, polymerase chain reactions (PCRs) were performed, targeting a ~434 bp fragment of the standard barcode region (Hebert et al. 2003) within the 5' portion of subunit I of the cytochrome c oxidase (COI) mitochondrial gene. Custom primers were designed to match sequences in aphids (target species: *R. maidis*, *R. padi*, *D. noxia*, *Myzus persicae*) but not ladybeetles (non-target species: *H. variegata*, *H. conformis*, *Coccinella transversalis*), to ensure target specificity. Partial Illumina adapter sequences (in bold italic) were added as primer tails at the 5' end of the aphid-specific primers, Aphid_F: 5'-***ACACTCTTTCCCTACACGACGCTCTTCCGATCT***TAAGACAAATTAATTCAATTATYAA-3'; Aphid_R: 5' - ***GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT***GGGAAAAGTGGAATTTGRTT-3'. The PCR was conducted in 25 µL reactions consisting of 14.7 µL Bovine Serum Albumin

(New England Biolabs, USA), 5 µL of 5× Bioline MyFi reaction buffer (Meridian Bioscience, USA), 1 µL of each primer (10 µM), 0.8 µL MyFi DNA polymerase (Meridian Bioscience, USA) and 2.5 µL of template DNA. The PCR cycling conditions were as follows: 2 minutes at 94 °C; 30 cycles of denaturation at 94 °C for 30 seconds, annealing at 49 °C for 45 seconds, and extension at 72 °C for 45 seconds; and then a final extension phase of 1 minute at 72 °C. Amplification was verified by gel electrophoresis (1% w/v agarose). After the initial PCR amplification, the amplicons were diluted 1/10 and run through a second 6 cycles of real-time PCR to attach unique dual indexes and the remainder of the Illumina adapter sequence. Each indexing real-time PCR reaction (50 µL volume) contained 32.5 µL of ddH_2O, 10 µL of 5× Phusion HF Buffer (Thermo Fisher Scientific, USA), 1 µL of dNTP mix (10 mM), 1 µL of SYBR Green I Mix (Thermo Fisher Scientific, USA) diluted 1/1000 in ddH_2O, 0.5 µL Phusion DNA polymerase (Thermo Fisher Scientific, USA), 4 µL of sample-specific indexing primers (2.5 µM) and 1 µL of the diluted PCR product. The real-time PCR cycling conditions were: 98 °C for 30 seconds, followed by 6 cycles of 98 °C for 10 seconds, 65 °C for 30 s, and 72 °C for 30 seconds, with fluorescence measurement conducted in the 72 °C phase. The amplification curve was visually inspected in real time and stopped while still in the exponential phase, during the extension phase of the 6th cycle, to prevent over-amplification artefacts. Amplicons were purified and normalised using the SequalPrep Normalization Plate Kit (Thermo Fisher Scientific, USA) following the manufacturer's protocol, but eluting the final product in 12 µL instead of 20 µL. Normalised and cleaned real-time PCR amplicons were then pooled together and the resulting library was quality checked, sized and quantified using a High Sensitivity D1000 ScreenTape assay performed on a 2200 TapeStation (Agilent Technologies, USA). The final pooled library was diluted to a concentration of 7 pM, spiked with 15% PhiX, and sequenced using V3 chemistry (2 × 250 bp reads) across two flow cells on an Illumina MiSeq system (Illumina, USA). Raw sequence reads were demultiplexed using bcl2fastq, allowing for a single mismatch in the indexes, and the FASTQ files were archived at NCBI SRA BioProject PRJNA1267727.

Bioinformatic analysis

Analysis of FASTQ files was performed using the Freyr metabarcoding analysis pipeline (commit 9d4f34a) available here: https://github.com/AVR-biosecurity-bioinformatics/freyr. Freyr was run with the parameters file "loci_params.csv" (Suppl. material 4), along with the command-line parameters "--primer_n_trim --miseq_internal". Briefly, reads per sample were grouped by their primer sequence (allowing a single mismatch) and then trimmed of primers, using Cutadapt (Martin 2011). Reads were then filtered using DADA2 (Callahan et al. 2016), retaining reads with <=1 expected error, a minimum length of 20 bp, and no ambiguous bases. Reads were then truncated to 225 bp and denoised using DADA2 in a two-step process, using sequences from the first round as priors for the second round. Paired-end reads were merged and the resulting ASVs were filtered with a minimum length of 220 bp and a maximum length of 390 bp. Taxonomy was assigned to ASVs using IDTAXA (Murali et al. 2018) with the aphid COI reference database described below used as reference, with a confidence threshold of 60%.

The reference database of aphid COI sequences (Suppl. material 5) was generated using the Mimir DNA barcoding reference database curation pipeline (commit e2ad316) available here: https://github.com/AVR-biosecurity-bioinformatics/mimir. Mimir was run with the command-line parameters "--target_taxon 33385 --target_rank superfamily --marker COI --min_length 200 --genbank_fetch_size 5000 --add_root --remove_unclassified any_ranks --placeholder_as_unclassified --max_group_size 3 --train_idtaxa --idtaxa_max_iterations 1", and used the April 4th 2025 data package from the BOLD database (Ratnasingham et al. 2024). Briefly, all Aphidoidea COI sequences and whole mitochondrial genomes were downloaded from GenBank and BOLD and subset to the COI coding region using a profile hidden Markov model from InterPro (Blum et al. 2025). These sequences were filtered to remove those with ambiguous bases and any unclassified taxonomic ranks, then clustered at a threshold of 97% and pruned of sequences that taxonomically disagreed with >=80% of their cluster at the genus level. Remaining sequences were grouped by species, identical sequences removed (leaving one per species), remaining sequences were aligned and those with genetic distances > 5% to the median sequence within each species were removed. Up to 3 sequences were retained per species, and an IDTAXA model was trained using a single iteration.

To further analyse the metabarcoding data in R, an index switching "noise floor" per flow cell was calculated from the estimated rate of index switching multiplied by the total number of reads retained in the Freyr output; ASVs with fewer reads than the noise floor had their read count set to 0 for subsequent analyses, on a per-sample basis. ASVs were removed if they were not assigned to the family Aphididae, with the exception of a number of 226 bp ASVs that had high sequence identity to COI from *Uroleucon* spp. but had stop codons and frameshifts. As these appeared in samples where *Uroleucon* spp. were expected to appear, we interpreted these sequences as nuclear mitochondrial (NUMT) COI pseudogenes from *Uroleucon* spp. and merged their read counts with all other *Uroleucon* sequences. ASVs with the same species assignment were merged into single species, with their read counts summed. Raw output and "cleaned" data files are available as Suppl. materials 6, 7, respectively.

Statistical analysis

Binomial regression modelling of aphid DNA detection predicted by sampling time in hours was performed using the glm() function in R ("logit" link function), with the predictor $\log_2(x+1)$ transformed, and the number of aphids consumed by each beetle (0–5) used as a sample weight. For each sample, "detection" was defined as a non-zero number of reads assigned to the fed aphid species. The 50% effective time (ET_{50}) -- time taken after feeding to reduce detection probability to 50% -- for each fed aphid experiment was determined from the regression model using the "dose.p()" function in the MASS package and back-transforming to the unit of hours.

Results

After quality control, 12.16 million reads (mean 95,790 ± se 10,040 per sample) were retained for subsequent analysis from a total of 127 samples.

Control samples

Our negative controls (beetles that were not provided with aphids during the feeding experiments) showed no reads assigned to any aphid species, except for *R. maidis* in the Short starvation experiment, which is the prey species the beetles were maintained on prior to the feeding experiment. Our positive controls (each aphid species alone) showed only reads for their own species, while psyllids (fed to some beetles in feeding experiments) produced no reads assigned to aphids. Finally, our technical controls (extraction blanks), showed no reads for aphids, apart from slight contamination in one extraction blank (Suppl. materials 1, 2).

Short starvation experiment

In the Short starvation experiment, *R. maidis* (consumed *ad libitum* prior to the feeding period) and *D. noxia* (consumed during the feeding period) aphids remained detectable at all measured time points up to and including 72 hours post-feeding (except for a sample at 4 hr where no *D. noxia* aphids were consumed). Overall, the number and proportion of reads attributable to *Uroleucon* spp. (consumed during the feeding period) was generally lower compared to either *R. maidis* or *D. noxia*. Specifically, *Uroleucon* spp. DNA was broadly detectable up to 4 hr post-feeding, but detectability was reduced at 72 hr post-feeding, with only one of the six replicates producing *Uroleucon* spp. reads (Fig. 2).

Long starvation experiment

In the Long starvation experiment, *R. maidis* and *B. brassicae* (consumed during feeding period) were consistently detected at 0 hr and 1 hr post-feeding, but detectability decreased in the 2 hr and 4 hr samples for both species. After 24 hr, aphid DNA was largely undetected, apart from a single replicate at 48 hr for *R. maidis* (Fig. 3). We detected *R. maidis* between 0 hr and 4 hr in beetles that were given *B. brassicae*, although at lower reads compared to the beetles fed on *R. maidis*. Although initial pilots for non-destructive extractions gave inconsistent PCR results (above), the non-destructively extracted samples in the Long starvation experiment showed qualitatively similar metabarcoding results to the destructively extracted samples (Suppl. material 2).

Model results for aphid detection as a function of time since feeding

The presence/absence of DNA from aphids fed during the feeding periods in the Short starvation experiment and Long starvation experiment, predicted by time since feeding (transformed as $\log_2(\text{hr} + 1)$), was modelled using logistic regression to produce "50% effective time" (ET_{50}) estimates, or the time after which the detection probability was 50% (Fig. 4). For *D. noxia* in the Short starvation experiment, no model could be fitted as detection occurred in nearly all samples (Fig. 4A), but significant models were produced for *Uroleucon* spp. in the Short starvation experiment ($p = 5.7 \times 10^{-5}$; Fig. 4B), *B. brassicae* in the Long starvation experiment ($p = 0.029$; Fig. 4C), and *R. maidis* in the Long starvation experiment ($p = 5.6 \times 10^{-9}$; Fig. 4D). Back-transformed ET_{50} values were 14.0 hours for *D. noxia* in the

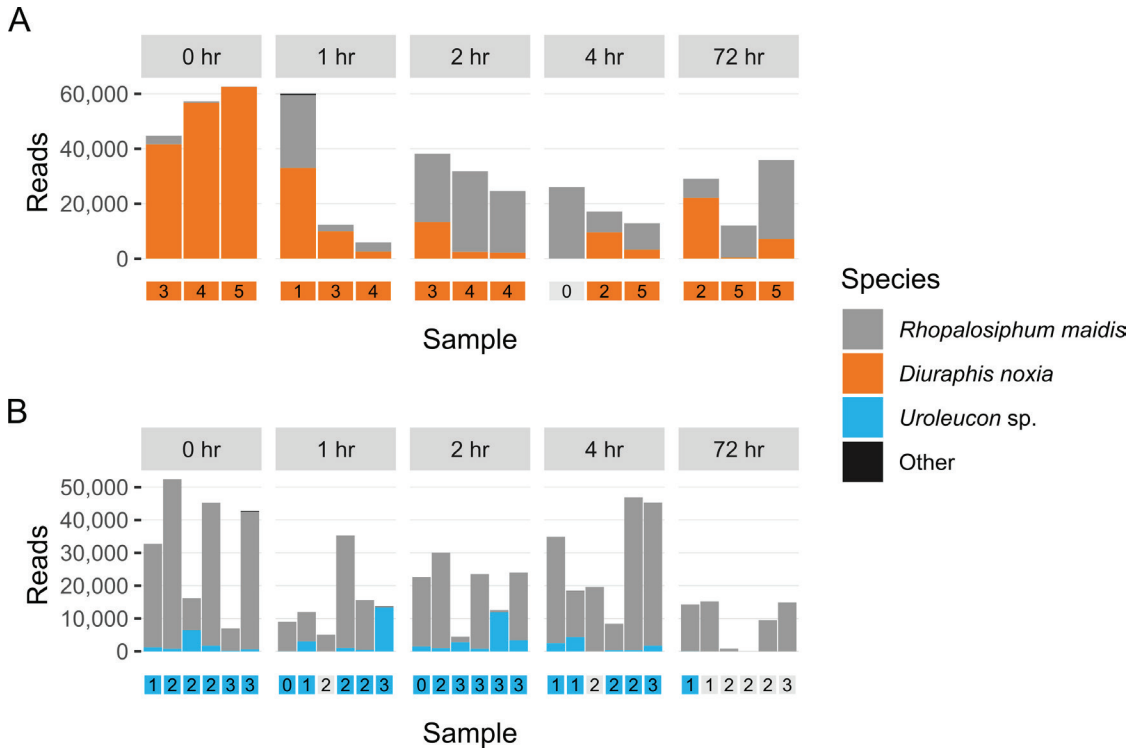


Figure 2. Results from the Short starvation experiment, for **A.** *Diuraphis noxia* and **B.** *Uroleucon* spp. samples. Each sample is displayed as a stacked bar of sequence reads, the colour of which conveys their taxonomic assignment. The box beneath each bar displays the number of target aphids consumed by the beetle (text) as well as the presence/absence of target aphid reads (colour). Samples are grouped horizontally by the sampling time post-feeding on the target species.

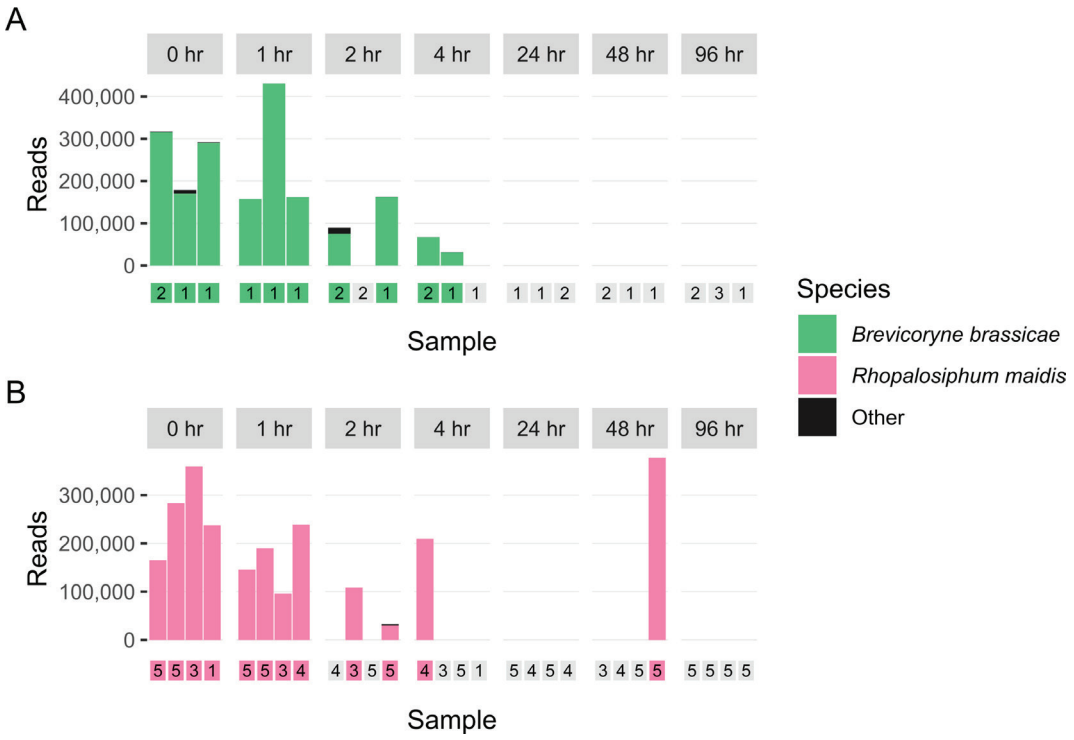


Figure 3. Results from the Long starvation experiment, for **A.** *Brevicoryne brassicae* and **B.** *Rhopalosiphum maidis* samples. Each sample is displayed as a stacked bar of sequence reads, the colour of which conveys their taxonomic assignment. The box beneath each bar displays the number of target aphids consumed by the beetle (text) as well as the presence/absence of target aphid reads (colour). Samples are grouped horizontally by the sampling time post-feeding on the target species.

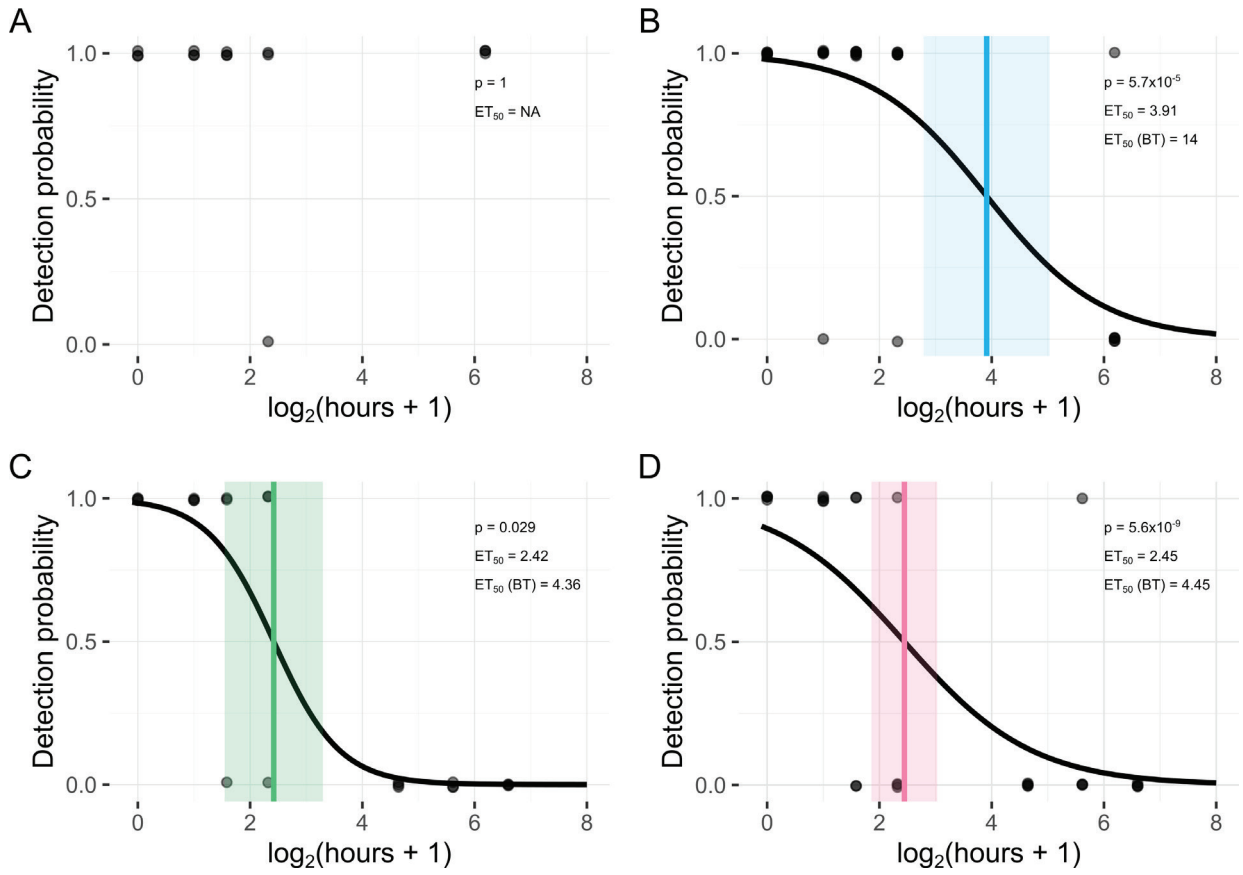


Figure 4. Logistic regression of aphid DNA detection predicted by the time since feeding, using number of consumed aphids as sample weights, for the Short starvation (**A, B**) and Long starvation (**C, D**) experiments. Dots are individual samples, with 0 and 1 indicating no detection and detection, respectively; y-axis values are slightly jittered to reduce overplotting. p, ET_{50} and back-transformed ET_{50} values are printed for each model where available; ET_{50} and its 95% CI is displayed as a coloured line and area. **A.** *Diuraphis noxia*; **B.** *Uroleucon* spp.; **C.** *Brevicoryne brassicae*; **D.** *Rhopalosiphum maidis*.

Short starvation experiment and 4.36 and 4.45 hours for *B. brassicae* and *R. maidis* in the Long starvation experiment, respectively, indicating a consistent decrease in the detectability window for aphid DNA from beetles starved for 96 hr compared with those starved for 17 hr.

Ladybeetle frass

Our preliminary data show that target aphids are also detectable in ladybeetle frass collected after, but not before, a feeding event (Suppl. material 1). For example, ladybeetle Hc1 was fed a *Uroleucon* spp. aphid, and this aphid was detected in frass produced after (Hc1P2), but not before (Hc1P1), the feeding event. Similarly, ladybeetle Hc5 showed a reading for the aphid *Macrosiphum corydalis* after, but not before, feeding. Ladybeetle Hv1, which was provided *Uroleucon* spp. during the feeding event, showed the same pattern, but also showed readings for other aphid species – *B. brassicae*, *M. corydalis*, *Macrosiphum euphorbiae*, and *Macrosiphum melampyri* (none of which were provided during the feeding event) – in frass from before feeding, and these other aphid species were not detected in frass collected on the following day after the feeding event.

Field samples

Specimens of *H. conformis* collected from wild populations produced reads for multiple aphid species. All field specimens showed reads assigned to *R. maidis*, and field-collected larvae also showed reads assigned to *R. padi*, *Hyperomyzus carduellinus*, and *Uroleucon* spp. (Suppl. material 1).

Discussion

The metabarcoding approach on which we based our study has been widely used for arthropod surveillance and diagnostics (Batovska et al. 2018; Martoni et al. 2023a, 2023b; Martoni et al. 2024). Thanks to the design of target-specific primers, we modified this protocol to assess whether DNA was present inside the insect guts and if it could be detected at different time points post-feeding, which is particularly valuable in an agricultural context. As hypothesised, we found that aphid DNA extracted from ladybeetles could be detected, but that this detectability declines over time in a species-specific way. We also found preliminary data demonstrating the efficacy of this technique for detecting aphid DNA in ladybeetle frass. These results match previous studies that have shown the utility of non-destructive insect metabarcoding for detecting the presence of insect pests and beneficials, as well as finding extra DNA reads not linked to morphological specimens present in the sample (Martoni et al. 2023a). Our results build on these findings, supporting the hypothesis that these reads can come from prey DNA in the guts of predator insects. This suggests that at least some metabarcoding results that are often classified as "false positives" (i.e. DNA is detected but the insect is not present; Darling et al. 2021) can be representative of the true presence of a target in the environment, even if the target's body is not visible in the sample.

Our aphid-specific primers detected all aphid species included in the experiments, following destructive and non-destructive DNA extractions from beetles. The detectability of aphids (based on the presence/absence of sequencing reads) differed between aphid species and generally declined with greater time between feeding and sampling of beetles. In addition to detecting aphids from whole insect extractions, we were able to detect aphids from trace material – ladybeetle frass – highlighting the value of this technique for environmental DNA (eDNA) studies.

The detectability periods of aphid DNA appeared to depend on the timing and quantity of aphids consumed. When ladybeetles were provided with large numbers of aphids with a short 17 hr starvation period (e.g. *R. maidis* in the Short starvation experiment), that aphid DNA extracted from ladybeetles remained consistently detectable up to at least 72 hours after feeding. By contrast, when few aphids were consumed with a longer 96 hr starvation period (e.g. *Uroleucon* spp. in the Short starvation experiment), detectability was reduced. In the Long starvation experiment, where there was a 96-hour starvation period prior to the feeding event, aphid DNA was not detected after 4 hours post-feeding. This means that aphids consumed in small amounts (*B. brassicae*, provided during the feeding event) had short detectability periods, whereas aphids consumed in large amounts (*R. maidis*, provided *ad libitum* prior to the feeding experiment) remained detectable at > 100 hours post-feeding. These patterns are consistent with previous studies, such as the DNA of a single *R. maidis* being detectable in

ladybeetles for approx. 15 hours (Chen et al. 2000), whereas *ad libitum* feeding leads to longer detectability half-lives (Greenstone et al. 2014). Interestingly, the detectability of target aphids lasted much longer in the Short starvation experiment (up to 72 hours) than in the Long starvation experiment (rarely more than four hours). This may reflect an effect of prior starvation increasing metabolic or digestive rate once feeding occurs and therefore producing faster DNA degradation (Weber and Lundgren 2009). This may have been exacerbated by the use of feeder psyllids in the Long starvation experiment maintaining this higher metabolic rate. Alternatively, ladybeetles in the Long starvation experiment may have consumed less DNA in total, and so were able to break it down faster, which would align with our hypothesis that starvation periods can impact DNA detectability. These potential impacts of metabolic rate on DNA detectability require further investigation. Nevertheless, our results highlight the importance of timing when sampling in agricultural settings, as well as the dynamics of each system. For example, in cases of early incursions of pests or predators into a new system, the insects collected may have undergone several days of starvation/low feeding. On the other hand, sampling in a system with high pest and predator abundance may result in a scenario more similar to a "no-starvation" experiment and pest detectability periods may be increased.

The differences in detectability of DNA between aphid species may also be driven by aphid traits. In the Short starvation experiment, we found more reads for *R. maidis* as explained above, but we also generally found a greater number of reads per sample for *D. noxia* compared with *Uroleucon* aphids, despite a limited number of each being provided to beetles at the start of the feeding trials. *Uroleucon* aphids have larger body sizes than *D. noxia*, and so beetles consuming them would be expected to ingest more DNA, increasing subsequent detectability (Greenstone et al. 2014). This would be concordant with previous research, which found that when the amount of DNA in the gut at time 0 (equivalent to our 0 hr) is greater, the DNA remains detectable for a longer period (Greenstone et al. 2014). However, *Uroleucon* sequences were not included in our aphid primer design, possibly explaining the lower number of reads assignable to these aphids. For the Long starvation experiment, the number of reads was generally comparable for the two target aphids, *B. brassicae* and *R. maidis* (except for one instance of *R. maidis* remaining detectable at 48 hours), likely because of the similarity in body size between the two aphid species. Thus, the effectiveness of surveillance may be impacted by traits such as prey body size. Furthermore, aphid body size, exoskeleton hardness, as well as primer bias across different species, are some of the well-known factors to bias the number of reads generated through metabarcoding analyses (Martoni et al. 2022).

Knowing detectability periods for pest DNA is vital for both designing and interpreting pest surveillance programs (Greenstone et al. 2014). Our metabarcoding approach also demonstrates the relative ease with which custom primers can be established to enrich for sequences from target species (aphids) consumed by non-target species (beetles). The results of our feeding experiments suggest that when ladybeetles have been feeding on high aphid densities, or perhaps have been feeding for longer, pest DNA remains detectable for longer. Conversely, if the pest species is newly established and/or at low densities, metabarcoding approaches may require more frequent ladybeetle sampling to capture these feeding events, as DNA detectability periods could be reduced.

The present study has some limitations that may impact its interpretations. As previously mentioned, our aphid-specific primers were not designed using *Uroleucon* sequences, which may have reduced their ability to detect species in this genus. We recommend that future works optimise these primers by designing them in the full context of known aphid COI sequences. Our two main feeding experiments used destructive DNA extractions, in order to improve the consistency of detecting prey DNA, but many metabarcoding workflows use non-destructive extractions to preserve insect specimens for morphological validation of species detections (Martoni et al. 2023a, 2023b). Our estimated ET_{50} values for aphid DNA detection after feeding are likely only directly applicable to destructive extraction-based metabarcoding; we suggest future work be conducted to estimate these values for non-destructive extractions. Ideally, our two starvation experiments would have also had identical, even and more densely sampled distributions of timepoints, which may have improved the precision of our ET_{50} estimates. Another limitation is that our experimental analysis of aphid DNA detection in ladybeetle frass was rudimentary, and a more robust time-course experiment would allow more quantitative conclusions to be drawn about the value of testing frass, especially in a field context. Additionally, our use of aphid-specific primers also prevented us from determining the relative abundance of prey DNA compared to predator DNA; such data could be used to explore the likelihood of detecting prey DNA from whole-community arthropod samples, which frequently use generic primers that amplify barcodes from a wide range of arthropod taxa (Piper et al. 2019).

Ultimately, this work highlights how HTS metabarcoding can record the presence of prey DNA from pools of insects, without the presence of physical specimens. In this context, this technique enables the detection of eDNA from the surrounding areas (i.e. wherever the predator could feed), providing valuable information on the ecological network present in a given environment. This provides an additional tool for surveillance and diagnostics, which would be unable to record prey presence using traditional morphological techniques if voucher specimens were missing, as hypothesised elsewhere (Martoni et al. 2023a). Additionally, this can also provide a tool for eDNA biodiversity assessment studies, where targeting predator species and/or their frass would enable obtaining information on the insect prey communities present in the environment. We note, however, that the use of target-specific primers enriches for target species relative to the rest of the ecological community, and, as such, conclusions drawn from subsequent metabarcoding data need to be interpreted with this in mind.

Our results demonstrate the ability of these metabarcoding approaches to detect prey DNA in predators and their frass, and we have established preliminary timeframes for detectability in this system. As hypothesised, DNA detectability was impacted by starvation periods and time since feeding. As such, we have also shown the need for these approaches to consider relationships between predator and prey densities, feeding frequencies and durations, and the factors that affect predator metabolic rates, all of which require further investigation. Nevertheless, this metabarcoding approach provides an efficient and effective tool for monitoring for the presence of pests and ensuring the effectiveness of biocontrol programs, as well as for better understanding the ecological network of insects present in agricultural crops.

Conclusion

Our study demonstrates that prey-targeted metabarcoding, enabled through the integration of aphid-specific primers into an established arthropod surveillance workflow, can reliably detect prey DNA within predatory insects and their frass across experimentally defined timeframes. Detection success was strongly influenced by prey species, feeding intensity, starvation history, and sampling timing, emphasising the importance of ecological context when interpreting metabarcoding data. These findings support the interpretation that unexpected prey detections in arthropod metabarcoding datasets may represent true environmental signals rather than methodological artefacts, often referred to as "false positive" results. By highlighting the role of primer design and feeding dynamics in shaping detectability, this work advances the application of metabarcoding for agricultural surveillance, biological control assessment, and eDNA-based monitoring, providing a foundation for refining sampling strategies and analytical frameworks in complex trophic systems.

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Additional information

Conflict of interest

The authors have declared that no competing interests exist.

Ethical statement

No ethical statement was reported.

Artificial Intelligence (AI) use

The authors accept full responsibility for the content of the manuscript, including the disclosure of any use of AI.
No AI tools were used in the preparation of this manuscript.
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Author contributions

Conceptualization: FM, NJW. Data curation: JLS, NN. Formal analysis: JLS, NJW. Funding acquisition: BCR, MJB. Investigation: FM, NJW, JLS. Methodology: NJW, FM, JLS. Project administration: BCR, FM, MJB, TMJ. Resources: NN. Software: JLS. Supervision: BCR, TMJ, FM, MJB. Visualization: JLS. Writing - original draft: JLS, FM, NJW. Writing - review and editing: MJB, NJW, JLS, FM, TMJ.

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Data availability

All of the data that support the findings of this study are available in the main text or Supplementary Information.

Supplementary material 1

Supplementary image 1

Authors: Nikolas J. Willmott, Jack L. Scanlan, Mark J. Blacket, Brendan C. Rodoni, Narelle Nancarrow, Therésa M. Jones, Francesco Martoni

Data type: pdf

Explanation note: Heatmap of (log10-transformed) read abundance per assigned species for samples in the Short starvation experiment FE flowcell L8GDJ. Samples are vertically facetted by type. Species with an untransformed abundance within a sample below the index switching noise floor of the flowcell (6 reads) have been set to 0 for that sample. Untransformed abundances of 0 are represented as unfilled tiles.

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Link: https://doi.org/10.3897/mbmg.10.196568.suppl1

Supplementary material 2

Supplementary image 2

Authors: Nikolas J. Willmott, Jack L. Scanlan, Mark J. Blacket, Brendan C. Rodoni, Narelle Nancarrow, Therésa M. Jones, Francesco Martoni

Data type: pdf

Explanation note: Heatmap of (log10-transformed) read abundance per assigned species for samples in the Long starvation experiment flowcell LD8N5. Samples are vertically facetted by type. Species with an untransformed abundance within a sample below the index switching noise floor of the flowcell (31 reads) have been set to 0 for that sample. Untransformed abundances of 0 are represented as unfilled tiles.

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Link: https://doi.org/10.3897/mbmg.10.196568.suppl2

Supplementary material 3

Supplementary table

Authors: Nikolas J. Willmott, Jack L. Scanlan, Mark J. Blacket, Brendan C. Rodoni, Narelle Nancarrow, Therésa M. Jones, Francesco Martoni

Data type: xlsx

Explanation note: Sample metadata including information on aphid specimens used in this work.

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Link: https://doi.org/10.3897/mbmg.10.196568.suppl3

Supplementary material 4

Supplementary file 1

Authors: Nikolas J. Willmott, Jack L. Scanlan, Mark J. Blacket, Brendan C. Rodoni, Narelle Nancarrow, Therésa M. Jones, Francesco Martoni

Data type: csv

Explanation note: Loci parameters file used to run the Freyr pipeline, supplied using the '—loci_params' argument.

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Link: https://doi.org/10.3897/mbmg.10.196568.suppl4

Supplementary material 5

Supplementary file 2

Authors: Nikolas J. Willmott, Jack L. Scanlan, Mark J. Blacket, Brendan C. Rodoni, Narelle Nancarrow, Therésa M. Jones, Francesco Martoni

Data type: fasta

Explanation note: COI reference database file for Aphidoidea generated by the Mimir pipeline and used for taxonomic assignment of sequencing reads using the Freyr pipeline.

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Link: https://doi.org/10.3897/mbmg.10.196568.suppl5

Supplementary material 6

Supplementary file 3

Authors: Nikolas J. Willmott, Jack L. Scanlan, Mark J. Blacket, Brendan C. Rodoni, Narelle Nancarrow, Therésa M. Jones, Francesco Martoni

Data type: csv

Explanation note: "Summary" output file of the Freyr pipeline for both flowcells. Each row is a sequence with its taxonomic assignment and read abundance per sample.

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Link: https://doi.org/10.3897/mbmg.10.196568.suppl6

Supplementary material 7

Supplementary file 4

Authors: Nikolas J. Willmott, Jack L. Scanlan, Mark J. Blacket, Brendan C. Rodoni, Narelle Nancarrow, Therésa M. Jones, Francesco Martoni

Data type: xlsx

Explanation note: Processed data used for analyses, with one sheet per experiment/flowcell. Each row is an observation of the read abundance of a particular species for a given sample.

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Link: https://doi.org/10.3897/mbmg.10.196568.suppl7